

# LABORATORY OBSERVATION

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**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Experiment No.:** 8

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## 1. TITLE

**Demonstrate HTML5 Graphics and CSS3 Web Styling Techniques**

## 2. AIM

To demonstrate HTML5 graphics using the canvas element and to apply advanced CSS3 styling techniques such as gradients, transitions and animations.

## 3. OBJECTIVE

The objectives of this experiment are:

- To draw basic shapes using the HTML5 <canvas> element and JavaScript.
- To apply CSS3 gradients, border-radius and box-shadow for styling.
- To apply a CSS3 transition for an interactive hover effect.
- To apply a CSS3 keyframe animation for a continuously animated element.

## 4. THEORY

The HTML5 <canvas> element provides a drawable region on a webpage that is controlled using JavaScript through the Canvas 2D API, accessed via `getContext("2d")`. It can be used to draw shapes, lines, text and images programmatically.

CSS3 introduced advanced visual styling features such as `linear-gradient()` backgrounds for smooth colour transitions, `box-shadow` for depth, and `border-radius` for rounded corners.

The CSS3 transition property animates a property change smoothly over a given duration (e.g., on hover), while `@keyframes` combined with the animation property defines a repeating sequence of style changes, such as a pulsing effect.

## 5. ALGORITHM / PROCEDURE

1. Create an HTML file with a <canvas> element.
2. Get the 2D drawing context using `getContext("2d")`.
3. Draw a filled rectangle using `fillRect()`.
4. Draw a filled circle using `arc()` and `fill()`.

5. Draw a line using `moveTo()`, `lineTo()` and `stroke()`.
6. Add two styled boxes below the canvas using linear-gradient backgrounds.
7. Apply border-radius and box-shadow to the styled boxes.
8. Add a CSS3 transition so the boxes scale up smoothly on hover.
9. Add a circular element with a `@keyframes` pulse animation.
10. Save the file and open it in a browser to view the graphics and effects.

## 6. PROGRAM

### index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML5 Graphics & CSS3 Styling</title>
  <style>
    canvas { background: white; border: 1px solid #ccc; }
    .box-row { display: flex; justify-content: center; gap: 24px; margin-top:
30px; }
    .box {
      width: 120px; height: 120px; border-radius: 16px;
      box-shadow: 0 4px 10px rgba(0,0,0,0.25);
      transition: transform 0.3s ease;
    }
    .box:hover { transform: scale(1.12); }
    .gradient1 { background: linear-gradient(135deg, #667eea, #764ba2); }
    .gradient2 { background: linear-gradient(135deg, #f6d365, #fda085); }
    .pulse {
      width: 60px; height: 60px; border-radius: 50%;
      background: #e74c3c; margin: 40px auto 0;
      animation: pulse 1.5s infinite;
    }
    @keyframes pulse {
      0% { transform: scale(1); opacity: 1; }
      50% { transform: scale(1.35); opacity: 0.55; }
      100% { transform: scale(1); opacity: 1; }
    }
  </style>
</head>
<body>
  <h1>HTML5 Graphics & CSS3 Styling Demo</h1>

  <canvas id="myCanvas" width="300" height="200"></canvas>

  <div class="box-row">
    <div class="box gradient1"></div>
    <div class="box gradient2"></div>
  </div>

  <div class="pulse"></div>

  <script>
    const canvas = document.getElementById("myCanvas");
    const ctx = canvas.getContext("2d");

    ctx.fillStyle = "#3498db";
    ctx.fillRect(20, 20, 100, 60);
```

```

    ctx.beginPath();
    ctx.arc(220, 50, 40, 0, Math.PI * 2);
    ctx.fillStyle = "#2ecc71";
    ctx.fill();

    ctx.beginPath();
    ctx.moveTo(20, 150);
    ctx.lineTo(280, 150);
    ctx.strokeStyle = "#e74c3c";
    ctx.lineWidth = 4;
    ctx.stroke();
</script>
</body>
</html>

```

## 7. CODE EXPLANATION

| Tag / Code                    | Explanation                                                       |
|-------------------------------|-------------------------------------------------------------------|
| <canvas id="">                | Creates a drawable canvas area on the page                        |
| getContext("2d")              | Returns the 2D drawing context used to draw on the canvas         |
| fillRect() / arc() / stroke() | Canvas methods used to draw a rectangle, a circle and a line      |
| linear-gradient()             | CSS3 function that creates a smooth colour gradient background    |
| border-radius / box-shadow    | CSS3 properties that round corners and add a shadow to an element |
| transition: transform 0.3s    | Smoothly animates the scale change when the box is hovered        |
| @keyframes pulse / animation  | Defines and applies a repeating scale + opacity animation         |

## 8. TEST CASES AND EXECUTION DATA

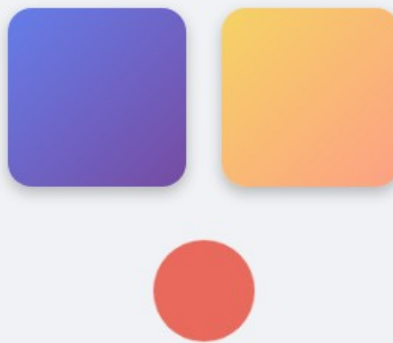
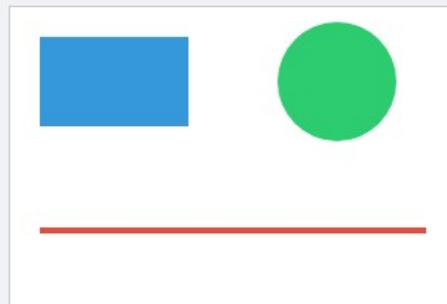
The page was opened in a browser and each graphic and styling effect was verified.

| Test Case | Description                  | Expected Result                                | Actual Result                      | Status |
|-----------|------------------------------|------------------------------------------------|------------------------------------|--------|
| TC-01     | Open index.html in a browser | Canvas shows a rectangle, circle and line      | Shapes displayed correctly         | Pass   |
| TC-02     | Hover over a gradient box    | Box scales up smoothly                         | Box scaled up smoothly             | Pass   |
| TC-03     | Observe the pulse circle     | Circle continuously pulses in size and opacity | Circle pulsed continuously         | Pass   |
| TC-04     | Check gradient boxes         | Boxes display smooth diagonal colour gradients | Gradients displayed correctly      | Pass   |
| TC-05     | Check box shadow and corners | Boxes show rounded corners and a soft shadow   | Rounded corners and shadow visible | Pass   |

### Test Case Output

**TC-01** — Canvas shows a rectangle, circle and line

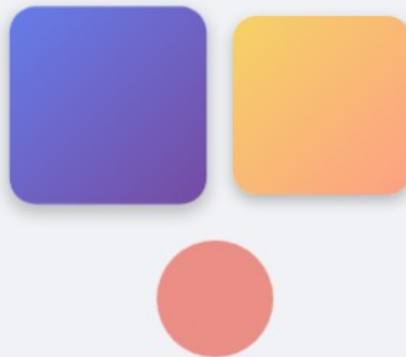
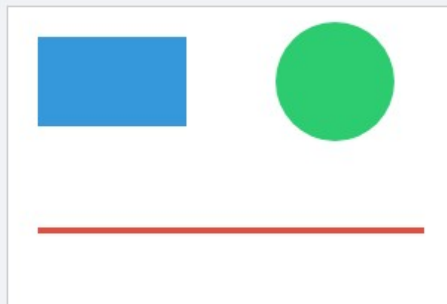
## HTML5 Graphics & CSS3 Styling Demo



*Canvas drawing showing the filled rectangle, circle and line*

**TC-02** — *Box scales up smoothly on hover*

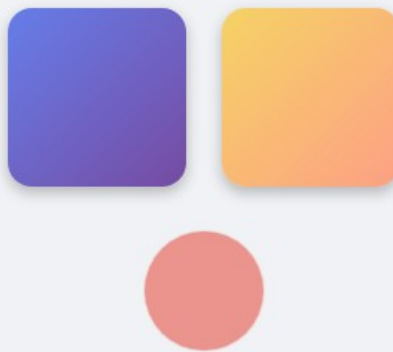
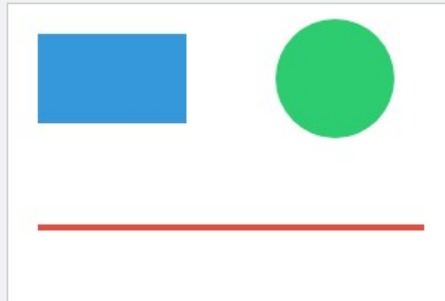
## HTML5 Graphics & CSS3 Styling Demo



*Purple gradient box scaled up (transform: scale) while hovered*

**TC-03** — Circle continuously pulses in size and opacity

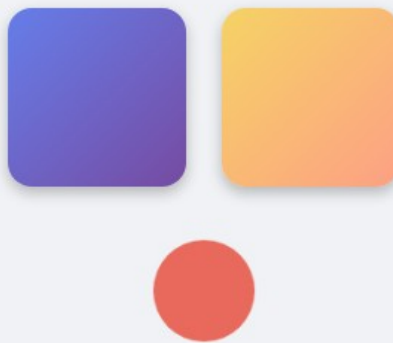
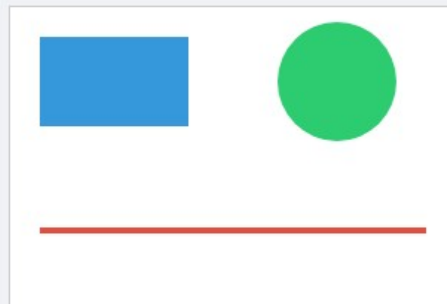
## HTML5 Graphics & CSS3 Styling Demo



*Circle captured mid-animation — enlarged and faded, confirming the keyframe animation is running*

**TC-04** — *Boxes display smooth diagonal colour gradients*

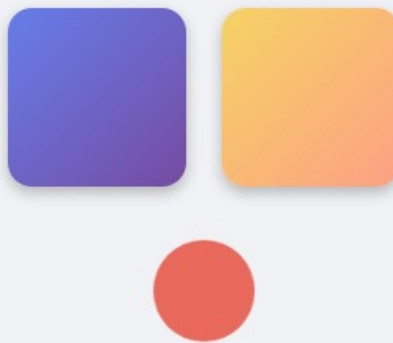
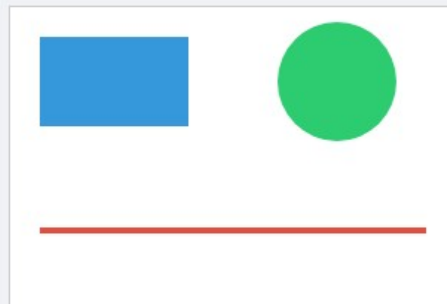
## HTML5 Graphics & CSS3 Styling Demo



*Purple-to-violet and yellow-to-orange gradients rendered on the two boxes*

**TC-05** — Boxes show rounded corners and a soft shadow

## HTML5 Graphics & CSS3 Styling Demo

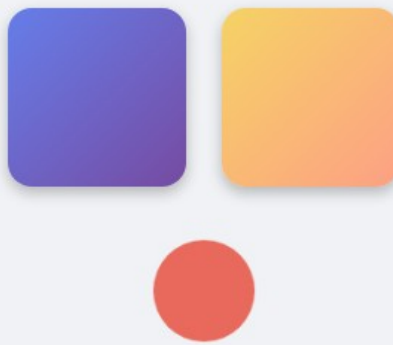
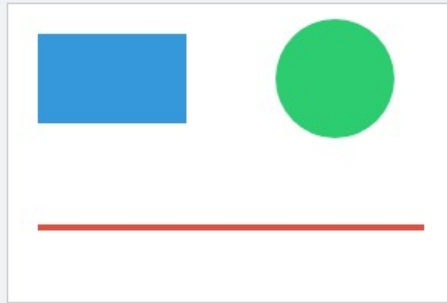


*border-radius and box-shadow visible on both gradient boxes*

### 9. OUTPUT

Output: Canvas Graphics and CSS3 Styling in Browser

## HTML5 Graphics & CSS3 Styling Demo



*Canvas shapes, gradient boxes and pulsing circle in their default state*

### 10. RESULT

Thus, HTML5 canvas graphics were successfully drawn and CSS3 styling techniques such as gradients, transitions and animations were successfully applied, and the output was verified in a web browser.